EdgeTrim-KD: Teacher-Guided Graph Sparsification for Efficient GNN Inference

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1 Description of the Problem

GNN inference cost grows with the number of edges, because every layer aggregates over each node's neighbors. Edge sparsification removes edges to cut that cost: joint edge-and-weight pruning [1, 4], one-shot degree-based pruning [2], and node-adaptive sparsification [3] already exist. Pruning edges by a fixed score is therefore not a contribution on its own.

Research question. Can a full-graph teacher GNN decide which edges to keep, and can distilling its soft predictions into a student trained on the sparse graph hold accuracy at a lower edge budget and a lower measured inference cost?

Approach: EdgeTrim-KD

1. Train a 2-layer GCN or GraphSAGE teacher [11, 12] on the full graph.
2. Score each edge by how much the teacher's predictions depend on it, for example the gradient of the teacher's loss with respect to a per-edge weight (one backward pass), and keep the highest-scoring edges under a budget (40–60% removed).
3. Train a student on the sparse graph with standard distillation [6], where p_T and p_S are the temperature-T softened teacher and student outputs:

$$\mathcal{L} = (1-\alpha)\mathcal{L}_{\text{CE}} + \alpha T^2 D_{\text{KL}}\left(p_T \,\|\, p_S\right). \tag{1}$$

Hypotheses.

- **H1:** Teacher-guided selection beats random, degree-based, and feature-similarity pruning at the same edge budget.
- **H2:** Distillation recovers accuracy that pruning alone loses.
- **H3:** The time and memory spent scoring edges are repaid by net inference savings.

2 Description of the Data

Two public node-classification benchmarks: Cora as a quick implementation check and ogbn-arxiv as the main benchmark (large enough for edge counts to move latency and memory), with the same models and splits for every pruning method; ogbn-products is a stretch dataset.

Metrics. Test accuracy (the OGB metric [10]), fraction of edges kept, inference latency, peak GPU memory, and edge-scoring cost, averaged over several seeds.

Dataset	Nodes	Edges	Features	Classes	Train / val / test
Cora [11]	2,708	5,429	1,433	7	140 / 500 / 1,000
ogbn-arxiv [10]	169,343	1,166,243 (directed)	128	40	By year: ≤ 2017 / 2018 / ≥ 2019

3 Plan Toward Completion

1. **Week 1: Baselines.** Set up Cora and ogbn-arxiv; train the full-graph teacher; implement random, degree-based, and feature-similarity pruning; record accuracy, latency, and peak memory with 0%, 20%, 40%, and 60% of edges removed.
2. **Week 2: Method.** Implement teacher-guided edge scoring and the distillation loss; run all five variants (the three baselines, teacher-guided, and teacher-guided plus distillation) at each edge budget over several seeds.
3. **Week 3: Analysis and write-up.** Ablate α and T; profile GPU latency and memory; plot accuracy against latency (Pareto), counting the scoring cost; write the report.

References

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